

RAMAN REVEALED

Advanced Data Analysis 1 (ADA) Map data



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Summary

- What is Multivariate statistical analysis?
- Comparing Component analysis and Empty Modelling™
- Component analysis
 - Direct classical and non-negative
 - Normalisation or no normalisation
 - Quantification using Concentration estimates
- Particle statistics
- Empty Modelling™ (MCR-ALS)

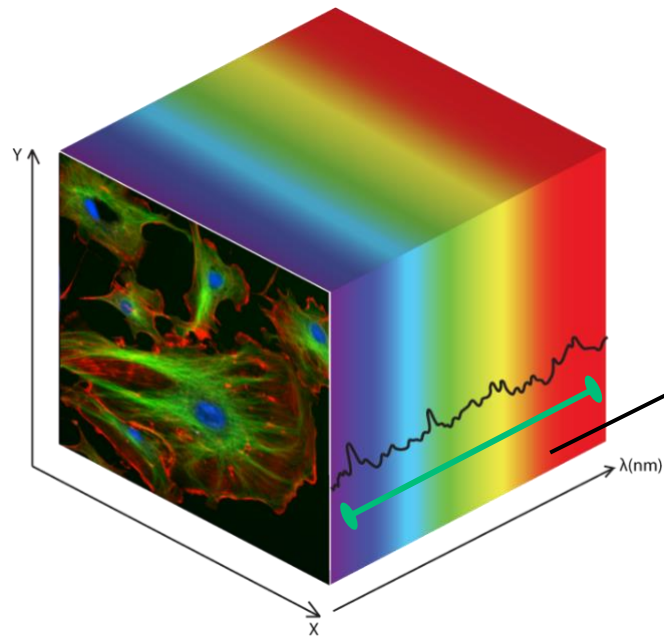
Introduction to multivariate analysis

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Multivariate statistical analysis – What is it?

- Multivariate statistical analysis
 - Analysis based on multiple variables
 - For Raman map data, enables images to be created based on the information from multiple Raman bands
- Supervised and unsupervised methods
 - Supervised, e.g. Component analysis (NNLS)
 - Unsupervised, e.g. Empty Modelling™ and Principal component analysis (PCA)

Common element (TM015)



For a given pixel point in the hyperspectral cube

Raw data = Model + Noise

The mathematical model:

$$X = T_a P'_a + E_a$$

Component analysis	Empty Modelling	PCA
T_a = concentration P_a = reference spectra E_a = the residuals P_a are given from a reference set	T_a = concentrations P_a = discovered spectra E_a = the residuals P_a are linearly independent and constrained to (real) positive	T_a = scores (concentration-like) P_a = loadings (spectrum-like) E_a = the residuals P_a are orthogonal and normalized

When to use which method

	Component analysis	Empty Modelling
What information is provided	Images representing: • spectra of the same shape in data (normalised - qualitative) • spectra of the same shape in data considering intensity changes (no normalisation - semi-quantitative)	• Reference 'spectra' • Identification • Images representing spectra of the same shape in data considering (normalised - qualitative)
Reference spectra/ standards	Required (potentially for all materials and correctly scaled for quantification requirements)	Not required
What is required	-	The main component should be present in pure form somewhere within the data
Essential pre-processing	-	• Removal of any spectral artefacts

Comparing Component analysis and Empty Modelling™

Component analysis	Empty Modelling™
When to use? • Reference spectra are available • To see an image of a specific material either in a qualitative or quantitative form • To look for the presence of a material • To find unknowns	When to use? • No reference spectra are available • Produce (pure) spectra representing real and physically meaningful components, and their corresponding images

Component analysis (NNLS, DCLS)

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Component analysis – least squares fitting (TM015)

For a given spectrum:

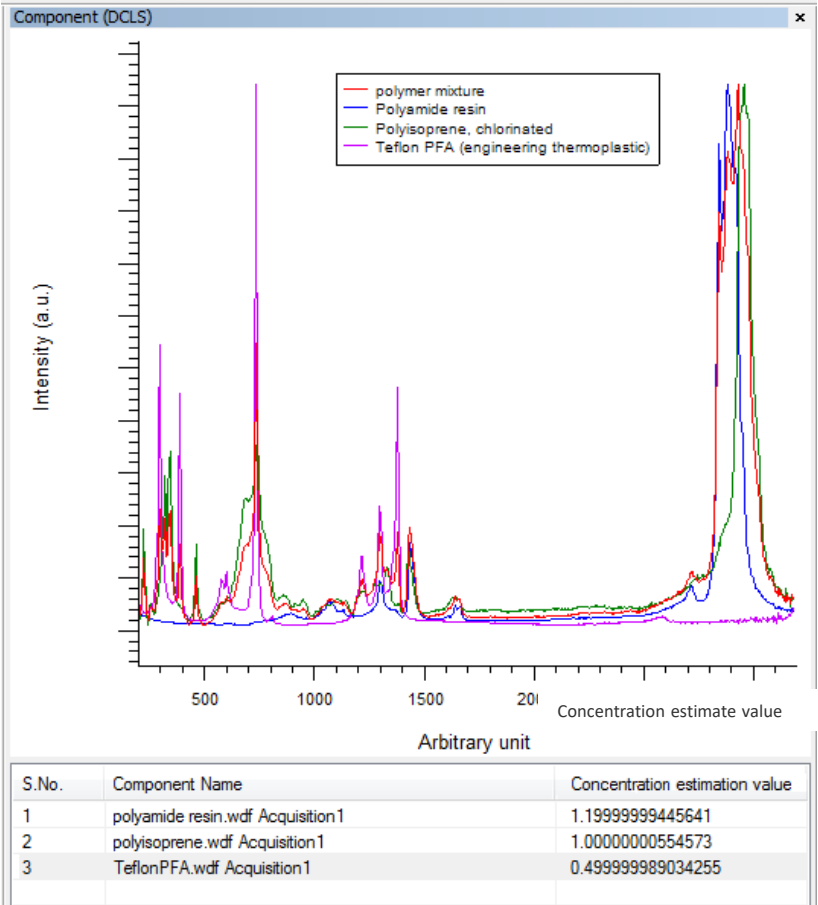
- Direct Classical Least Squares (DCLS) solves the least squares minimisation problem directly, without further constraints. This is very fast, but may produce negative values for the scaling factors derived for components that are not present in the spectrum. The benefit of NNLS is that it prevents non-physical negative values being obtained for the scaling factors. The mathematical solution must be achieved iteratively, and so is potentially slightly slower than DCLS.
 - Reference Spectrum 1 + Reference Spectrum 2 etc... + lack of fit = **measured spectrum**
 - If reference spectra are **not** collected under equivalent conditions, **normalisation is needed**
 - If reference spectra are available for a limited number of components, we can still run component analysis, however the concentrations will be inaccurate. The DCLS or NNLS 'lack of fit' map can potentially be used to identify chemical components which have been omitted or are unexpected

Component analysis – least squares fitting for a single spectrum

- **Component analysis (least squares)** can be used on a single spectrum to determine the concentration estimate (or concentration) of known species within a mixture.
- Component analysis uses a least squares approach, and requires reference spectra of known materials as an input. The reference spectra should be acquired using identical measurement conditions (e.g. laser wavelength, laser power, exposure time, accumulations) to enable quantitative information.

Test spectrum and input
reference spectra

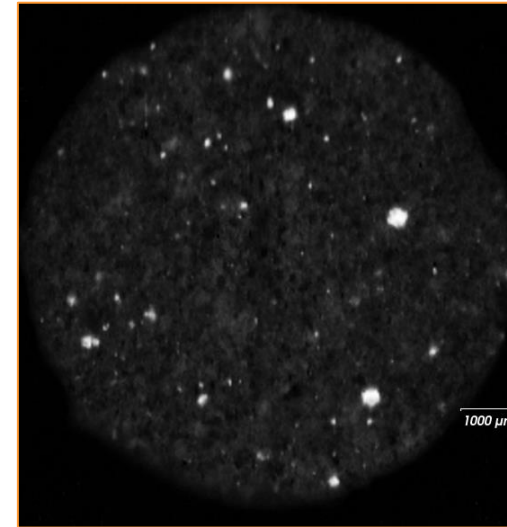
Component
analysis
concentration
estimate value



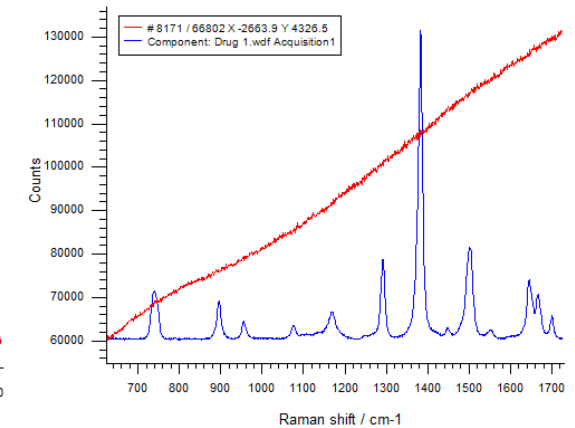
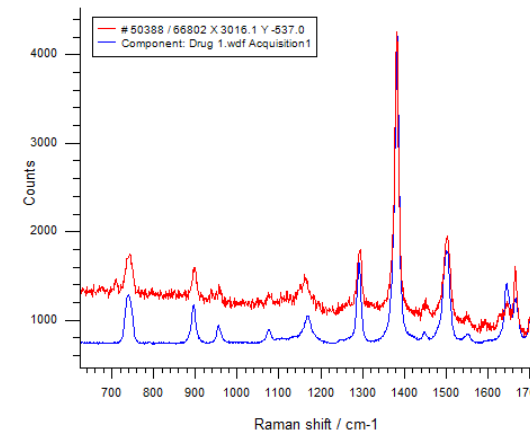
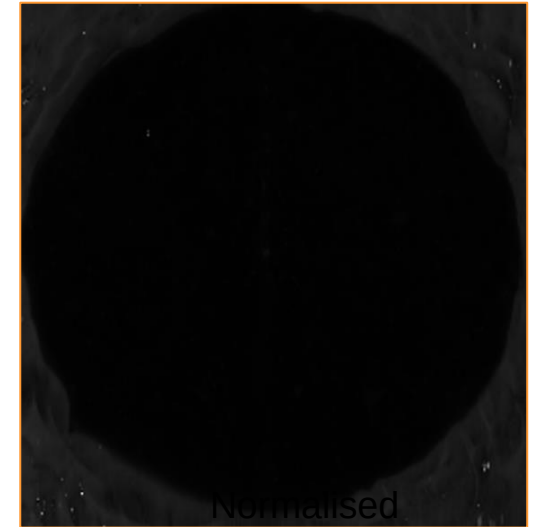
Component analysis – Normalisation

- **Normalisation**
 - Qualitative
- **No-normalisation**
 - Quantitative
- Tablet containing two drugs
- Aim: Produce concentration images for each drug
- Normalisation - images show good specificity but aren't quantitative
- No normalisation - Surface contaminants/surrounding substrate produce high background data which skews concentration images and values
- No normalisation with masking/baseline subtraction – High quality concentration images

Normalised



Not normalised



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Component analysis

Component Analysis Properties

Analysis

Components

Type

☐ Direct Classical Least Squares (DCLS)

☒ Non-Negative Least Squares (NNLS)

Method

☒ Spectrum

☐ 1st Derivative

☐ 2nd Derivative

Normalise

☐ No normalisation

☒ Mean centre and scale to unit variance

☐ Subtract minimum intensity and scale to unit intensity sum

Differential

Scale (num.data points)

3

☒ Auto-set

Polynomial order

2

☒ Savitsky-Golay

Background

☐ Include background

Polynomial order

0

☐ Calculate percentage lack of fit

OK

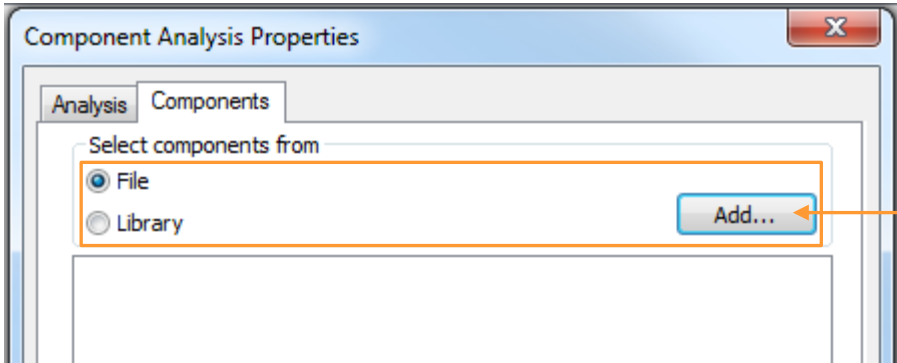
Cancel

Apply

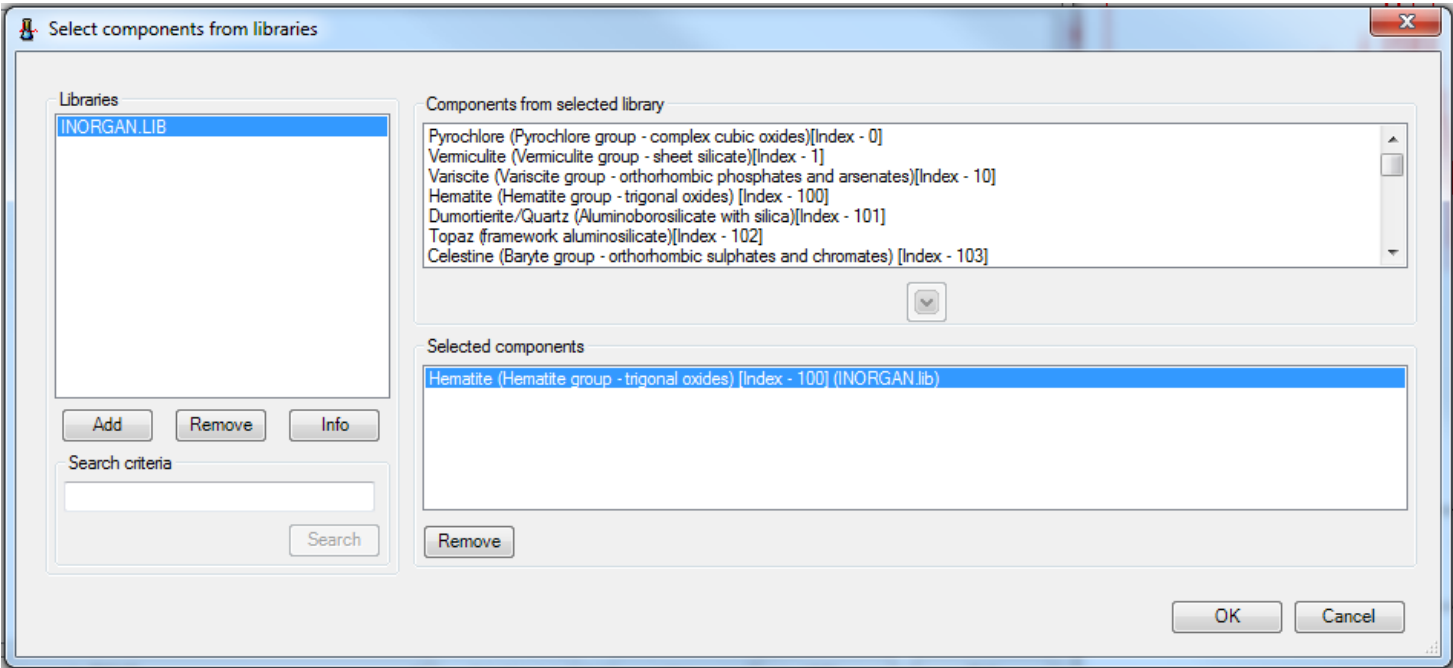
Help

Select the 'Components' tab to add reference spectra.

Component analysis



Components can be selected from a file or from a spectral library



Component analysis

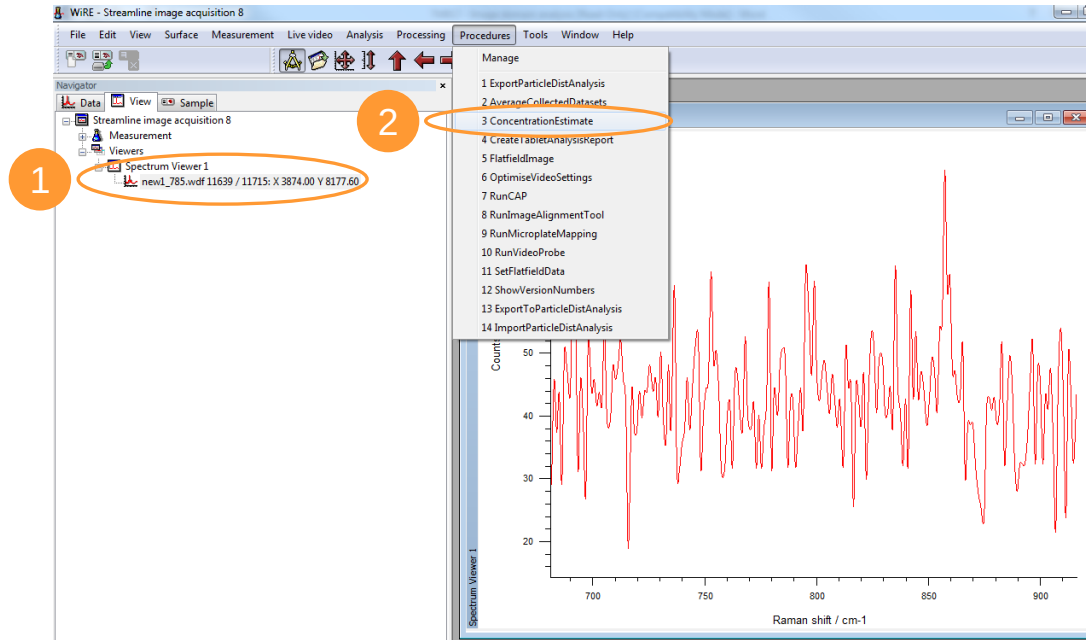
- 1 Choose a component analysis method
- 3 Choose a method of normalisation:
 - a. Normalise if absolute intensities are not important (**Mean centre and scale to unit variance** is set as default).
 - b. Choose **No normalisation** if absolute intensity variations are required (quantitative information)
- 5 Use **Calculate percentage lack of fit** to create an additional image highlighting regions which deviate most from the optimum solution (i.e. spectral information is not well accounted for by the reference spectra)
 - This can be used to identify chemical components which have been omitted from the references or are unexpected

The screenshot shows the 'Component Analysis Properties' dialog box with the 'Analysis' tab selected. The 'Type' section has 'Non-Negative Least Squares (NNLS)' selected. The 'Method' section has 'Spectrum' selected. The 'Normalise' section has 'Mean centre and scale to unit variance' selected. The 'Differential' section has 'Scale (num. data points)' set to 3, 'Polynomial order' set to 2, and both 'Auto-set' and 'Savitsky-Golay' checked. The 'Background' section has 'Include background' checked and 'Polynomial order' set to 0. The 'Calculate percentage lack of fit' checkbox is also checked. Arrows from the numbered text boxes point to these specific settings.

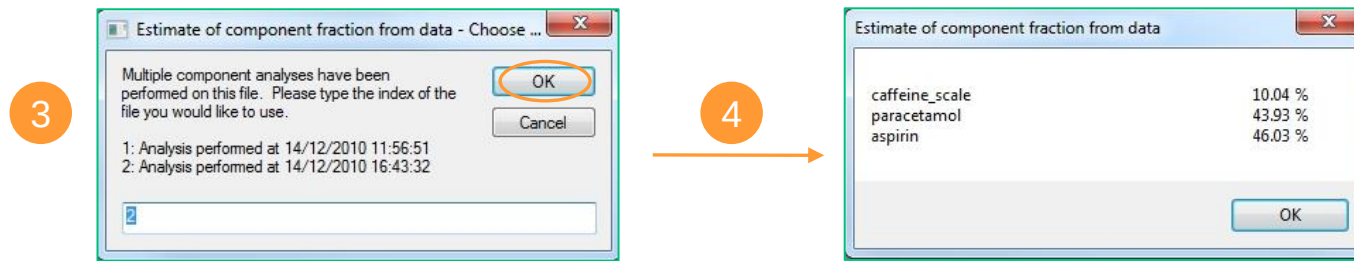
- 2 Choose **Spectrum** if a direct comparison is required and the signal to noise is not very high, especially where absolute intensity values are important
 - a. Choose **1st Derivative** if the data quality is very high and the user wishes to remove consistent background information
 - b. Choose **2nd Derivative** if, in addition to the 1st differential requirements, the background changes dynamically (e.g. fluorescence quenching)
- 4 Select the background option (only available for Spectrum method) if there is variable background in the map data, to be modelled during the analysis

Component analysis – Concentration estimates

- The Concentration estimate procedure calculates the component analysis 'concentration' values
- Adjusting the LUT and displayed image has no effect on the Concentration estimate result



1. Ensure the data is open in a Spectrum Viewer window (not Map Review).
2. Select **Procedures > ConcentrationEstimate**
3. Concentration estimates can be determined for each instance of NNLS/DCLS analysis.
4. The data fraction (%) results for each component will then be displayed.



Exercise ADA1 – 1. Qualitative and quantitative Component analysis

- Aims:
 - Produce Raman image showing the presence of caffeine only
 - Use Concentration estimates to quantify the relative amount of aspirin, paracetamol and caffeine
- Data file(s):
 - 'Tablet.wdf' is a map data file acquired from an internal (milled) surface of an analgesic tablet
 - Three component reference spectra have been acquired under equivalent conditions:
 - 'aspirin.wdf'
 - 'caffeine.wdf'
 - 'paracetamol.wdf'

Exercise ADA1 – 1. Qualitative and quantitative Component analysis

- Aim:
 - Produce Raman image showing the presence of Caffeine only
- Method:
 1. Create a NNLS Raman image with **normalisation** showing the distribution of the caffeine component

To create the NNLS map

 2. Open 'tablet.wdf' in a new window within WiRE
 3. Create a NNLS map using the 'caffeine.wdf'
 - Click on **Analysis > Mapping Review**
 - Select **Component analysis (least squares)** from the drop-down menu, and click on **Create**
 - In the Component analysis dialog, select **NNLS** and **Spectrum** as the analysis type and method
 - **Include background** with a polynomial order of 4. Use the default **Normalisation** option
 - Click **OK** to create the map
 4. Select and view the map to see the caffeine particles. Click on the particles to check the map spectrum against the reference (validation that the image accurately represents the reference)

Exercise ADA1 – 1.Qualitative and quantitative Component analysis

- Aim:
 - Use Concentration estimates to quantify the relative amount of aspirin, paracetamol and caffeine
- Method:
 1. Create a NNLS Raman image with ***no normalisation*** showing the distribution of the aspirin, paracetamol and caffeine component

To create the NNLS map

2. Open 'tablet.wdf' in a new window within WiRE, if not already open
3. Create NNLS maps using the 'aspirin.wdf', 'paracetamol.wdf' and 'caffeine.wdf' files
 - Click on ***Analysis > Mapping Review***
 - Select ***Component analysis (least squares)*** from the drop-down menu, and click on ***Create***
 - In the Component analysis dialog, select ***NNLS*** and ***Spectrum*** as the analysis type and method
 - ***Include background*** with a polynomial order of 4. Use ***No normalisation***
 - Click ***OK*** to create the maps
4. Select and view the maps to validate spectra at bright pixels for each image correlate well to the relevant reference spectrum

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Exercise ADA1 – 1.Qualitative and quantitative Component analysis

- Aim:
 - Use Concentration estimates to quantify the relative amount of aspirin, paracetamol and caffeine
 - Method:
5. Close ***Mapping Review***
 6. Select ***Procedures....Concentration estimates***
 - Choose the last analysis performed and view the percentage values for each component

Exercise ADA1 – 2. Find unknown contaminants using NNLS lack of fit

- Aim:
 - To determine unexpected components and locate them using lack of fit
- Data file(s):
 - 'Photovoltaic.wdf' is a map data file acquired from a semiconductor sample
 - A component reference spectrum has been acquired which represents the main component present: 'main reference.wdf'
- Method:
 1. Create a NNLS map which shows the detection of contaminant using known references

To create the NNLS map

2. Open 'Photovoltaic.wdf' in a new window within WiRE

Exercise ADA1 – 2. Find unknown contaminants using NNLS lack of fit

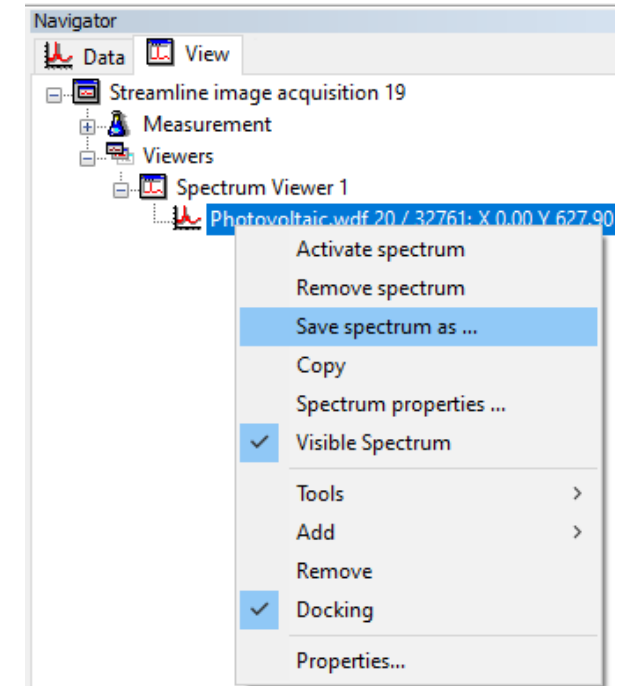
- Aim:
 - To determine unexpected components and locate them using lack of fit
- Method:
 3. Create a NNLS map using the component reference spectrum: 'main reference.wdf'
 - Click on ***Analysis > Mapping Review***
 - Select ***Component analysis (least squares)*** from the drop-down menu, and click on ***Create***.
 - In the Component analysis dialog, select ***NNLS*** and ***Spectrum*** as the analysis type.
 - Use the default settings for Method and Normalise
 - Select ***Include background*** with a polynomial order of 4
 - Select ***Calculate percentage lack of fit***
 4. Adjust the LUT and create an overlaid Raman image of 'main reference' and the 'lack of fit'

Exercise ADA1 – 2. Find unknown contaminants using NNLS lack of fit

- Aim:
 - To determine unexpected components and locate them using lack of fit
- Method:
 5. Create component spectra for any unexpected species using the 'lack of fit' map, and then recreate the NNLS map including the newly detected species

To obtain component spectra:

6. Place the crosshairs on any 'bright' regions in the 'lack of fit' map. Inspect and compare the test (sample) spectrum to the known references
7. If the test spectrum contains Raman bands that are unique (and **not** due to cosmic rays), then save the spectrum for later to use as a component reference
 - To do this, expand the Map Review branch in the Navigator View tab (*Navigator>View>Viewers>Map Review*). Right-click on the spectral acquisition to be saved and select **Save spectrum as ...** (see image)



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Exercise ADA1 – 2. Find unknown contaminants using NNLS lack of fit

- Aim:
 - To determine unexpected components and locate them using lack of fit

- Method:

To create the NNLS map of the unexpected components:

8. Recreate the NNLS map of unexpected components on cigs using the 'main reference.wdf' **and** the component references for the newly detected species. Use the 'lack of fit' map to ensure that no more unexpected species remain
9. If required, repeat until all real chemical components have been accounted for in the resulting NNLS map

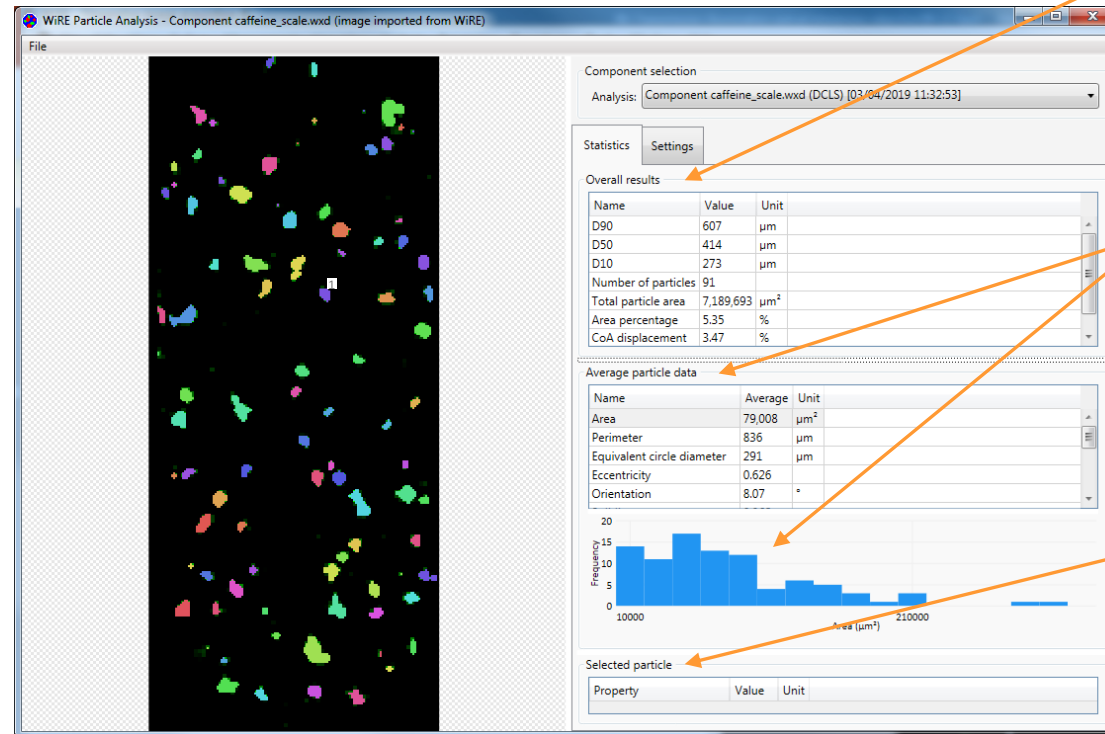
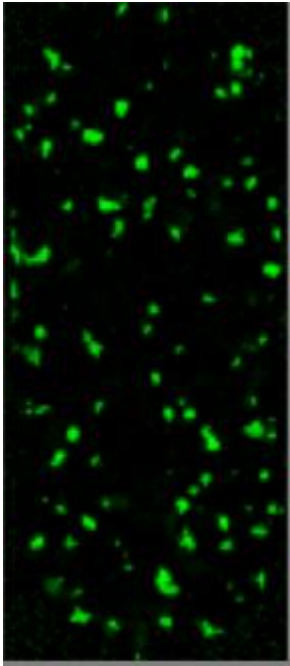
Particle statistics

- Particle statistics analyses an image and produces individual and average statistics on the particles/domains
- Adjusting the LUT affects the starting point of the particle statistics thresholding

- 1

- Particle statistics analyses the active image, and is sensitive to the LUT.
- 2

Procedures >
ExportToParticleDistAnalysis



Overall results are shown for the entire image.

Particle metrics in Average particle data can be selected to show the distribution of this metric in a histogram.

Individual particles can be selected
Properties of the selected particle are shown.

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Particle statistics

3

- Adjust settings to optimise the particle image

Binarisation Auto-binarisation uses Otsu's method to determine the image thresholding value. Untick 'Auto-binarise' to enable access to the slider which controls the threshold value as a percentage of the maximum image value.

Particle filtering Remove 'Edge particles' removes particles or domains which are touching the edges of the image. Particles can also be filtered using their size (% of largest particle).

Component selection

Analysis: Component_caffeine_scale.wxd (DCLS) [03/04/2019 11:32:53]

Statistics Settings

Reset to Default Settings

Image sampling

Resampling level 1

Interpolated (factor 4) from 11715 to 46389 points

☐ Invert brightness

Binarisation

☒ Auto-binarise

Manual binarisation value (% of maximum intensity) 30.20

Particle separation

Separation method ☐ Spatial ☒ Intensity

Valley depth (% of maximum intensity) 99.94

Particle filtering

☒ Particle filter

☐ Remove edge particles

Remove particles smaller than (µm) 0.00

Display

☐ Mono-coloured particles

☒ Overlay analysis results

☒ Show original image

☐ Show crosshair for centre-of-area

☐ Show crosshair for centre of particle image

Image sampling The sampling of the image (the number of image pixels analysed compared to the native number) can be manually set.

Particle separation Different separation methods (spatial or optimised) are provided independent of the binarisation step, each controlled as a percentage of the valley depth.

Display There are several options to affect the images which are displayed and how these are viewed.

4

- Selecting **File** then **Export results to text:** exports original LUT thresholding values, settings used, and all statistics values
- **Export particle values:** exports all individual domain parameter values.

Exercise ADA1 – 3. Quantify particle size and area using particle statistics

- Aim:
 - Quantify domain parameters of caffeine to show 'average equivalent circle diameter' and 'area percentage'
- Data file(s):
 - 'Tablet.wdf' is a map data file acquired from an internal (milled) surface of an analgesic tablet
 - 'caffeine.wdf' is a reference spectrum of caffeine
- Method:
 1. Create a NNLS Raman image with *normalisation* showing the distribution of the caffeine component

Exercise ADA1 – 3. Quantify particle size and area using particle statistics

- Method:

To create the NNLS map

2. Open 'Tablet.wdf' in a new window within WiRE

3. Create a NNLS map using 'caffeine.wdf'

- Click on **Analysis > Mapping Review**

- Select **Component analysis (least squares)** from the drop-down menu, and click on **Create** - In the Component analysis dialog, select **NNLS** and **Spectrum** as the analysis type and method

- *Include background* with a polynomial order of 4. Use the default normalisation option

- Click *OK* to create the map

4. Select and view the map to see the caffeine particles

5. To quantify the particle parameters:

- Click *Procedures > ExportToParticleDistAnalysis*.

- Adjust the settings so the particles in the Particle Statistics image match those in the Raman image

6. View the average equivalent circle diameter, and area percentage values.

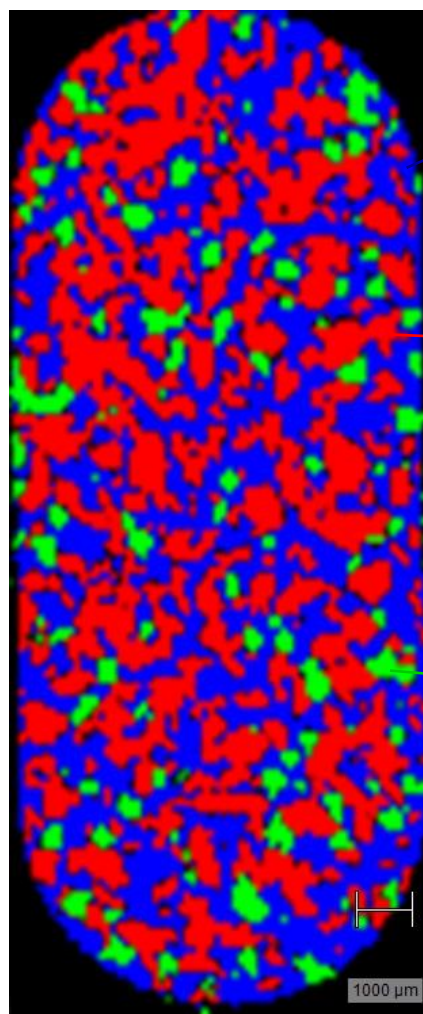
Empty Modelling™ (MCR-ALS)

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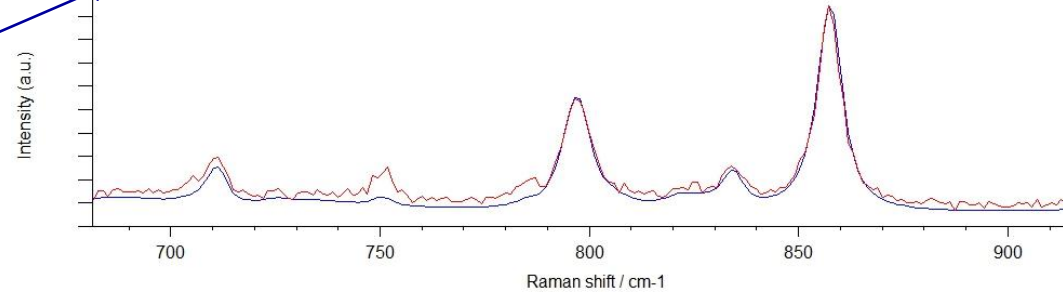
Empty Modelling™

- Look for a set of physically meaningful spectra (i.e. positive) which describes variations in the data
- Empty Modelling™ uses a “flat” assumption for each initial estimate of the spectra, and refines them iteratively using ALS to find the best description of the data
- The spectra are constrained to be positive, and (optionally) the concentrations are constrained to sum to 1 for each pixel
- New components are added until sufficient variance is explained in the data

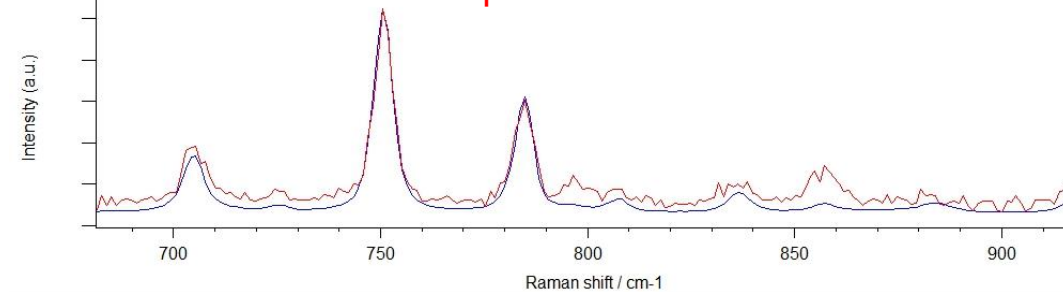
Empty Modelling™ – Example



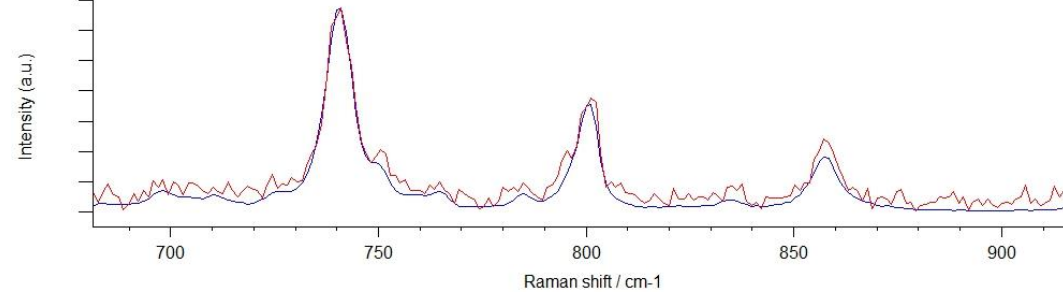
Resolved curve 1: Paracetamol



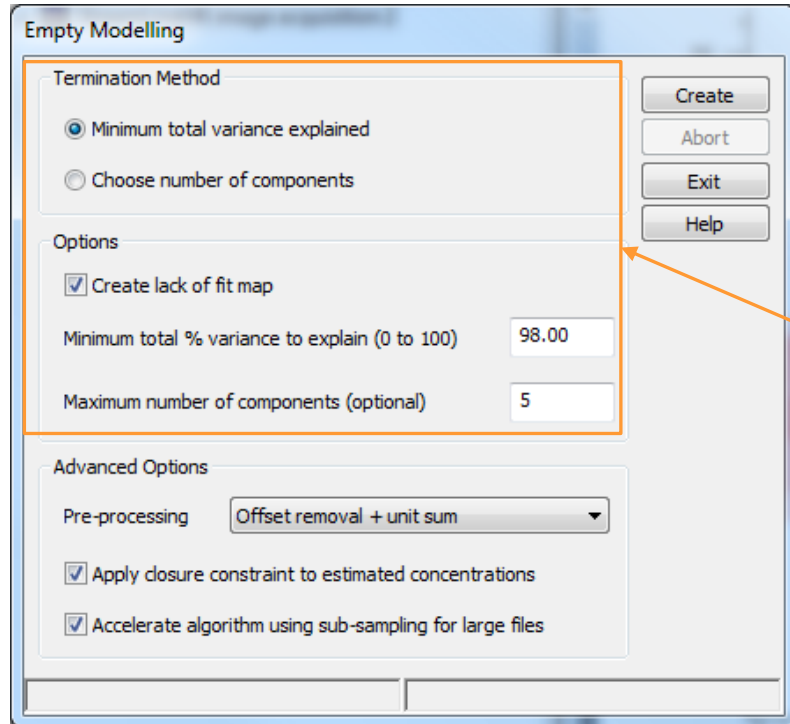
Resolved curve 2: Aspirin



Resolved curve 3: Caffeine



Empty Modelling™ - pre-processing options



The screenshot shows the 'Empty Modelling' dialog box. It has a title bar 'Empty Modelling'. Inside, there are three main sections: 'Termination Method', 'Options', and 'Advanced Options'. The 'Termination Method' section has two radio buttons: 'Minimum total variance explained' (selected) and 'Choose number of components'. The 'Options' section has a checked checkbox 'Create lack of fit map', a text input 'Minimum total % variance to explain (0 to 100)' with the value '98.00', and a text input 'Maximum number of components (optional)' with the value '5'. The 'Advanced Options' section has a 'Pre-processing' dropdown menu set to 'Offset removal + unit sum', and two checked checkboxes: 'Apply closure constraint to estimated concentrations' and 'Accelerate algorithm using sub-sampling for large files'. On the right side of the dialog, there are four buttons: 'Create', 'Abort', 'Exit', and 'Help'. An orange arrow points from a text box on the right to the 'Options' section of the dialog.

Empty Modelling

Termination Method

☒ Minimum total variance explained

☐ Choose number of components

Options

☒ Create lack of fit map

Minimum total % variance to explain (0 to 100) 98.00

Maximum number of components (optional) 5

Advanced Options

Pre-processing Offset removal + unit sum

☒ Apply closure constraint to estimated concentrations

☒ Accelerate algorithm using sub-sampling for large files

Create

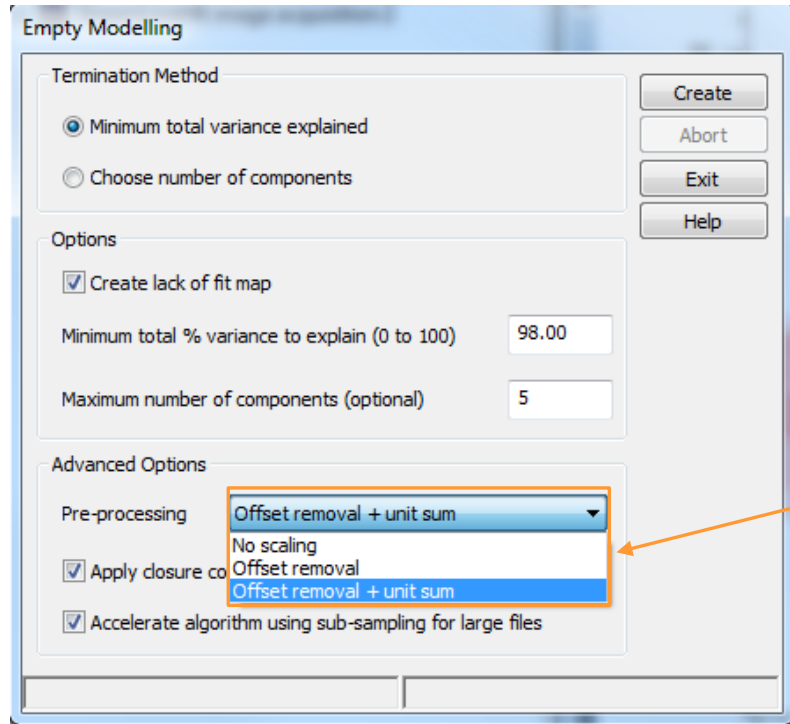
Abort

Exit

Help

- ***Decide termination method –***
- specify the minimum variance to explain as the output number
- Specify the number of components

Empty Modelling™ - pre-processing options



All the supplied scaling options operate on each spectrum individually.

- **Offset removal + unit sum** is the default option in WiRE. This option applies offset removal, and then scales each spectrum such that the sum of all points equals to 1. This may be a suitable option where maps contain a small amount of dead space but some normalisation is still desired.

Offset removal + Unit sum scaling focuses on “what” rather than “how much”.

- **No scaling** – to directly analyse the data without further scaling. This can be used to reflect all differences between the spectra in the file. As for DCLS/NNLS, this is useful for if there is a lot of noise in the background.

Exercise ADA1 – 4. Produce pure spectra as outputs from mixed map data

- Aim:
 - Produce individual spectra representing pure materials from tablet map data
- Data file(s):
 - 'Tablet.wdf' is a map data file acquired from an internal (milled) surface of an analgesic tablet
 - Three component reference spectra have been acquired under equivalent conditions:
 - 'aspirin.wdf'
 - 'caffeine.wdf'
 - 'paracetamol.wdf'
- Method:
 1. Create MCR-ALS maps for Anadin using the 'Tablet.wdf' map data file in WiRE, using the default settings.

Exercise ADA1 – 4. Produce pure spectra as outputs from mixed map data

- Aim:
 - Produce individual spectra representing pure materials from tablet map data
- Method:
 2. Inspect the loadings generated by MCR-ALS with the known reference spectra of aspirin, caffeine and paracetamol

To view the loadings and references in a new window. (**Navigator>Data**, expand the menu of the file, expand **Maps**, expand **Empty Modelling**, right click on **Empty Modelling component 1,2,3** etc and **Save loadings as...** Please type in “Component 1,2,3” etc in the file name to help distinguish the spectra later. Open the component curves by **File>Open** in new window.)